



MATSE: Operations Research

Lecture 8. Game Theory

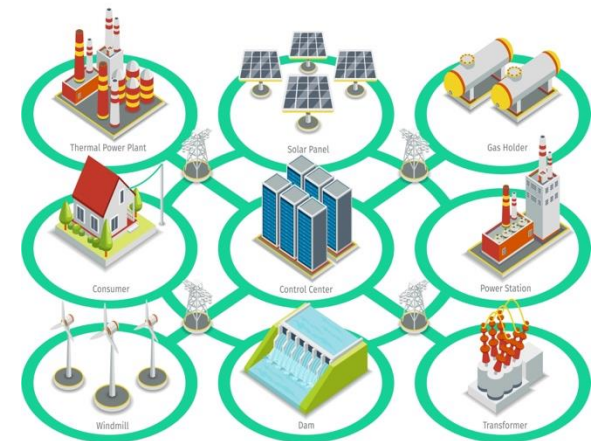
[based on slides and notes by: Bob Thuijsman, Steven Chaplick, Monica Salvioli]

What is Game Theory?

- Mathematical model for decision makers who interact
- Each of them tries to maximize its profit while influencing the others
- The choice made by one affects also the others interests

⇒ Cooperation, Competition, Conflicts

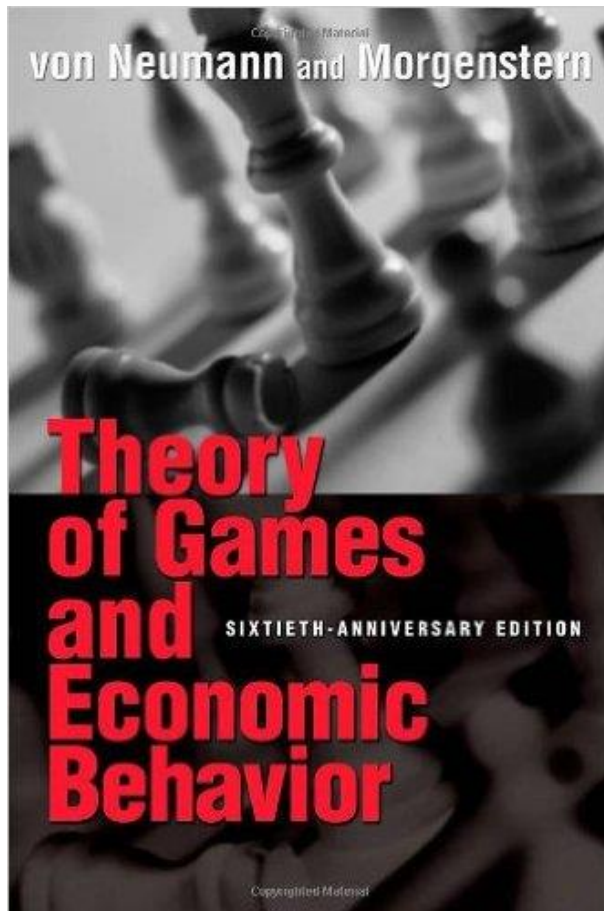
- Many applications



Images from [images.google.com](https://www.google.com) and [freepik.com](https://www.freepik.com)

Milestones

1944



- Formalization of the concept of game
- Winning strategies are independent of the others behaviour
- It was born during the cold war...

Milestones

Nash Jr, J. F. (1950). Equilibrium points in n -person games. *Proceedings of the national academy of sciences*, 36(1), 48-49.

This follows from the arguments used in a forthcoming paper.¹² It is proved by constructing an "abstract" mapping cylinder of λ and transcribing into algebraic terms the proof of the analogous theorem on CW-complexes.

* This note arose from consultations during the tenure of a John Simon Guggenheim Memorial Fellowship by MacLane.

¹ Whitehead, J. H. C., "Combinatorial Homotopy I and II," *Bull. A.M.S.*, 55, 214-245 and 453-496 (1949). We refer to these papers as CH I and CH II, respectively.

² By a complex we shall mean a connected CW complex, as defined in §5 of CH I. We do not restrict ourselves to finite complexes. A fixed 0-cell $e^0 \in K^0$ will be the base point for all the homotopy groups in K .

³ MacLane, S., "Cohomology Theory in Abstract Groups III," *Ann. Math.*, 50, 736-761 (1949), referred to as CT III.

⁴ An (unpublished) result like Theorem 1 for the homotopy type was obtained prior to these results by J. A. Zilber.

⁵ CT III uses in place of equation (2.4) the stronger hypothesis that λB contains the center of A , but all the relevant developments there apply under the weaker assumption (2.4).

⁶ Eilenberg, S., and MacLane, S., "Cohomology Theory in Abstract Groups II," *Ann. Math.*, 48, 326-341 (1947).

⁷ Eilenberg, S., and MacLane, S., "Determination of the Second Homology . . . by Means of Homotopy Invariants," these *PROCEEDINGS*, 32, 277-280 (1946).

⁸ Blikers, A. L., "Some Relations Between Homology and Homotopy Groups," *Ann. Math.*, 49, 428-461 (1948), §12.

⁹ The hypothesis of Theorem C, requiring that $\nu^{-1}(1)$ not be cyclic, can be readily realized by suitable choice of the free group X , but this hypothesis is not needed here (cf. *).

¹⁰ Eilenberg, S., and MacLane, S., "Homology of Spaces with Operators II," *Trans. A.M.S.*, 65, 49-99 (1949); referred to as HSO II.

¹¹ $C(K)$ here is the $C(K)$ of CH II. Note that $\bar{K}$ exists and is a CW complex by (N) of p. 231 of CH I and that $p^{-1}K^0 = \bar{K}^0$, where p is the projection $p: \bar{K} \rightarrow K$.

¹² Whitehead, J. H. C., "Simple Homotopy Types." If $W = 1$, Theorem 5 follows from (17:3) on p. 155 of S. Lefschetz, *Algebraic Topology*, (New York, 1942) and arguments in §6 of J. H. C. Whitehead, "On Simply Connected 4-Dimensional Polyhedra" (*Comm. Math. Helv.*, 22, 48-92 (1949)). However this proof cannot be generalized to the case $W \neq 1$.

EQUILIBRIUM POINTS IN N -PERSON GAMES

By JOHN F. NASH, JR.*

PRINCETON UNIVERSITY

Communicated by S. Lefschetz, November 16, 1949

One may define a concept of an n -person game in which each player has a finite set of pure strategies and in which a definite set of payments to the n players corresponds to each n -tuple of pure strategies, one strategy being taken for each player. For mixed strategies, which are probability

distributions over the pure strategies, the pay-off functions are the expectations of the players, thus becoming polylinear forms in the probabilities with which the various players play their various pure strategies.

Any n -tuple of strategies, one for each player, may be regarded as a point in the product space obtained by multiplying the n strategy spaces of the players. One such n -tuple counters another if the strategy of each player in the countering n -tuple yields the highest obtainable expectation for its player against the $n - 1$ strategies of the other players in the countered n -tuple. A self-countering n -tuple is called an equilibrium point.

The correspondence of each n -tuple with its set of countering n -tuples gives a one-to-many mapping of the product space into itself. From the definition of countering we see that the set of countering points of a point is convex. By using the continuity of the pay-off functions we see that the graph of the mapping is closed. The closedness is equivalent to saying: if $P_1, P_2, \dots$ and $Q_1, Q_2, \dots, Q_n, \dots$ are sequences of points in the product space where $Q_n \rightarrow Q$, $P_n \rightarrow P$ and Q_n counters P_n then Q counters P .

Since the graph is closed and since the image of each point under the mapping is convex, we infer from Kakutani's theorem¹ that the mapping has a fixed point (i.e., point contained in its image). Hence there is an equilibrium point.

In the two-person zero-sum case the "main theorem"² and the existence of an equilibrium point are equivalent. In this case any two equilibrium points lead to the same expectations for the players, but this need not occur in general.

* The author is indebted to Dr. David Gale for suggesting the use of Kakutani's theorem to simplify the proof and to the A. E. C. for financial support.

¹ Kakutani, S., *Duke Math. J.*, 8, 457-459 (1941).

² Von Neumann, J., and Morgenstern, O., *The Theory of Games and Economic Behaviour*, Chap. 3, Princeton University Press, Princeton, 1947.

REMARK ON WEYL'S NOTE "INEQUALITIES BETWEEN THE TWO KINDS OF EIGENVALUES OF A LINEAR TRANSFORMATION"*

By GEORGE POLYA

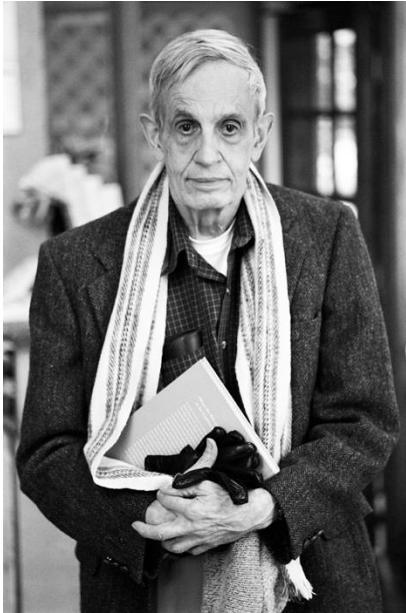
DEPARTMENT OF MATHEMATICS, STANFORD UNIVERSITY

Communicated by H. Weyl, November 25, 1949

In the note quoted above H. Weyl proved a Theorem involving a function $\varphi(\lambda)$ and concerning the eigenvalues α_i of a linear transformation A and those, κ_i , of A^*A . If the κ_i and $\lambda_i = |\alpha_i|^2$ are arranged in descending order,

Milestones

Nash Jr, J. F. (1950). Equilibrium points in n-person games. *Proceedings of the national academy of sciences*, 36(1), 48-49.



- Proposed a fundamental concept that will be known as Nash equilibrium
- Harsanyi-Nash-Salten Nobel 1994

Two other Nobel:

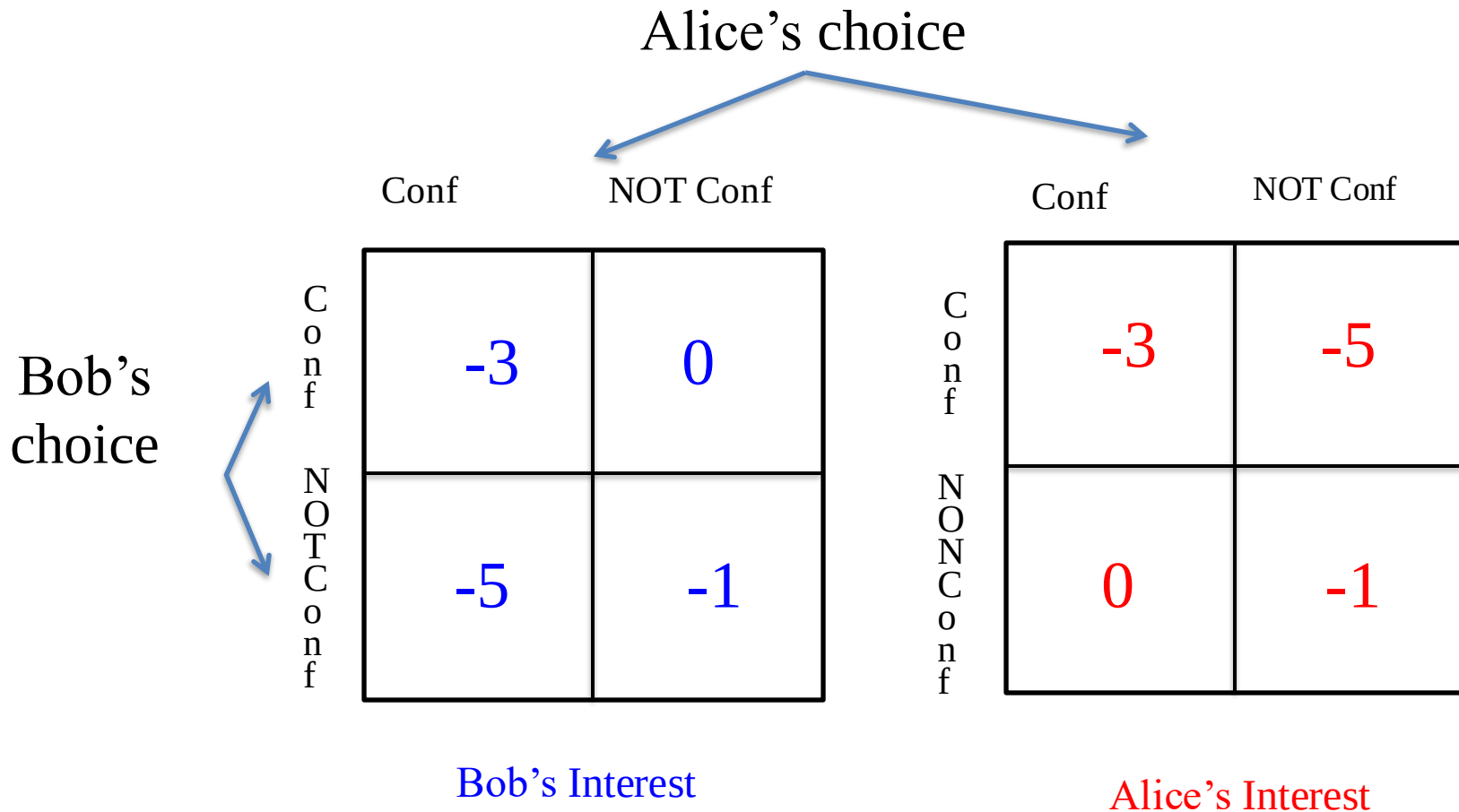
- 2005 Aumann-Shelling - Conflicts and Cooperation
- 2012 Roth-Shapley - Applications on Martkes

Prisoner's Dilemma

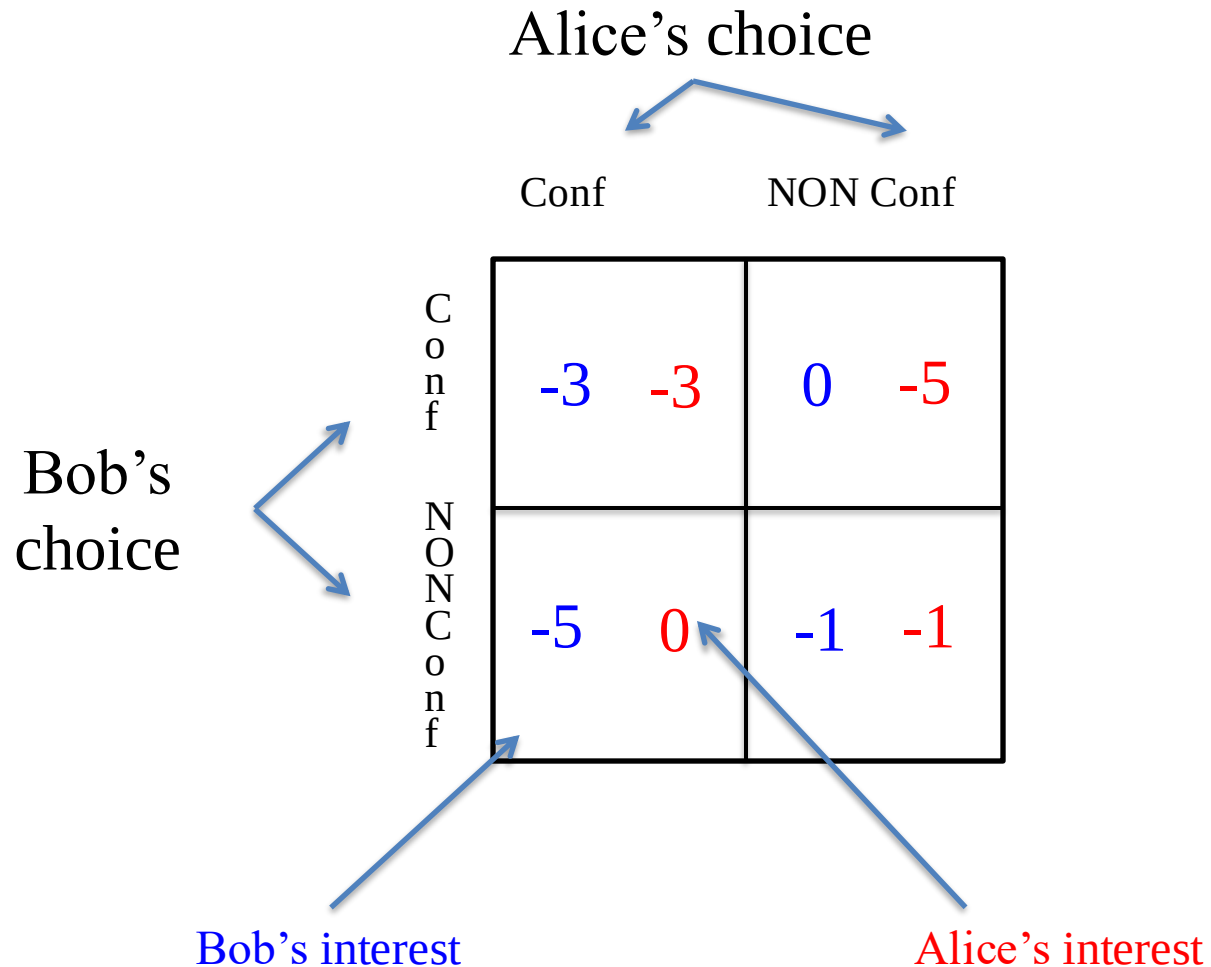
Bob and Alice have been arrested and accused of a crime. They are questioned separately (with no possibility of talking with each other) and they are proposed to Confess.

- If both Confess, both serve 3 years
- If one betrays the other, the first is free while the second serves 5 years
- If both don't Confess, they serve one year

Formalizing the game



Formalizing the game



Formalizing the game

Alice			
		Conf	NON Conf
Bob	Conf	-3 -3	0 -5
	NON Conf	-5 0	-1 -1

If Alice Confess, it is better for Bob to Confess.

Formalizing the game

		Alice	
		Conf	NON Conf
Bob	Conf	-3 -3	0 -5
	NON Conf	-5 0	-1 -1

If Alice doesn't Confess, it is better for Bob to Confess.

Formalizing the game

		Alice	
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Bob	Conf	-3 -3	0 -5
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It is always better for Bob to Confess!

Formalizing the game

		Alice	
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Bob	Conf	-3 -3	0 -5
	NON Conf	-5 0	-1 -1

It is always better for Bob to Confess!
Conf is a **dominant strategy** for Bob.

Formalizing the game

		Alice	
		Conf	NON Conf
Bob	Conf	-3 -3	0 -5
	NON Conf	-5 0	-1 -1

If Bob doesn't Confess, it is better for Alice to Confess.

Formalizing the game

		Alice	
		Conf	NON Conf
Bob	Conf	-3 -3	0 -5
	NON Conf	-5 0	-1 -1

If Bob doesn't Confess, it is better for Alice to Confess.

Conf is a **dominant strategy** for Alice.

Formalizing the game

		Alice	
		Conf	NON Conf
Bob	Conf	-3 -3	0 -5
	NON Conf	-5 0	-1 -1

Rational Choice: both Confess

Formalizing the game

Alice			
		Conf	NON Conf
Bob	Conf	-3 -3	0 -5
	NON Conf	-5 0	-1 -1

Rational Choice: both Confess

But it would be much better if they didn't!

Prisoner's Dilemma

Conf is a dominant strategy: rational choice

Implicit underlying assumptions:

- Bob and Alice cannot talk with each other
- We play the game only once
- The interest is determined only in the years they have to serve

Other possibilities:

- Repeated games
- Different interests for the players (altruism, fear...)

Altruistic Prisoner's Dilemma

Alice's choice

Conf NON Conf

Bob's choice

Conf

NON Conf

Conf	-3 -3	-2 -5
NON Conf	-5 -2	-1 -1

The matrix shows the following payoffs:

- If Alice chooses Conf and Bob chooses Conf: Alice gets -3, Bob gets -3.
- If Alice chooses Conf and Bob chooses NON Conf: Alice gets -2, Bob gets -5.
- If Alice chooses NON Conf and Bob chooses Conf: Alice gets -5, Bob gets -2.
- If Alice chooses NON Conf and Bob chooses NON Conf: Alice gets -1, Bob gets -1.

Altruistic Prisoner's Dilemma

Alice's choice

Conf NON Conf

Bob's choice

Conf

NON Conf

Conf	-3 -3	-2 -5
NON Conf	-5 -2	-1 -1

The matrix shows the following payoffs (Alice, Bob):

- If Alice Confesses and Bob Confesses: (-3, -3)
- If Alice Confesses and Bob does not Confess: (-2, -5)
- If Alice does not Confess and Bob Confesses: (-5, -2)
- If Alice does not Confess and Bob does not Confess: (-1, -1)

If Alice Confess, it is better for Bob to Confess.

If Alice doesn't Confess, it is better for Bob to not Confess.

Altruistic Prisoner's Dilemma

Alice's choice

Conf NON Conf

Bob's choice

Conf	-3 -3	-2 -5
NON Conf	-5 -2	-1 -1

If Alice Confess, it is better for Bob to Confess.

If Alice doesn't Confess, it is better for Bob to not Confess.

Altruistic Prisoner's Dilemma

Alice's choice

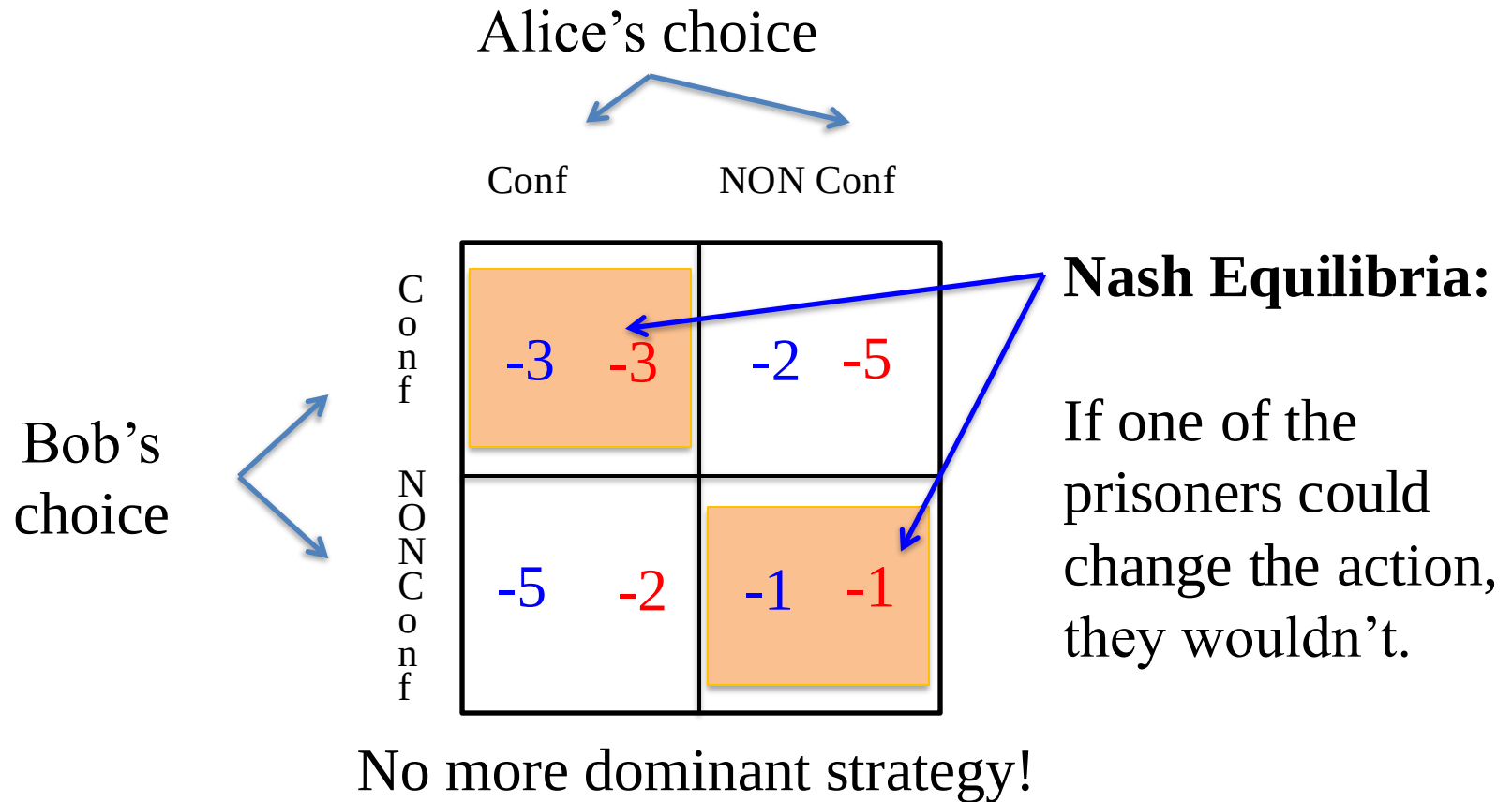
Conf NON Conf

Bob's choice

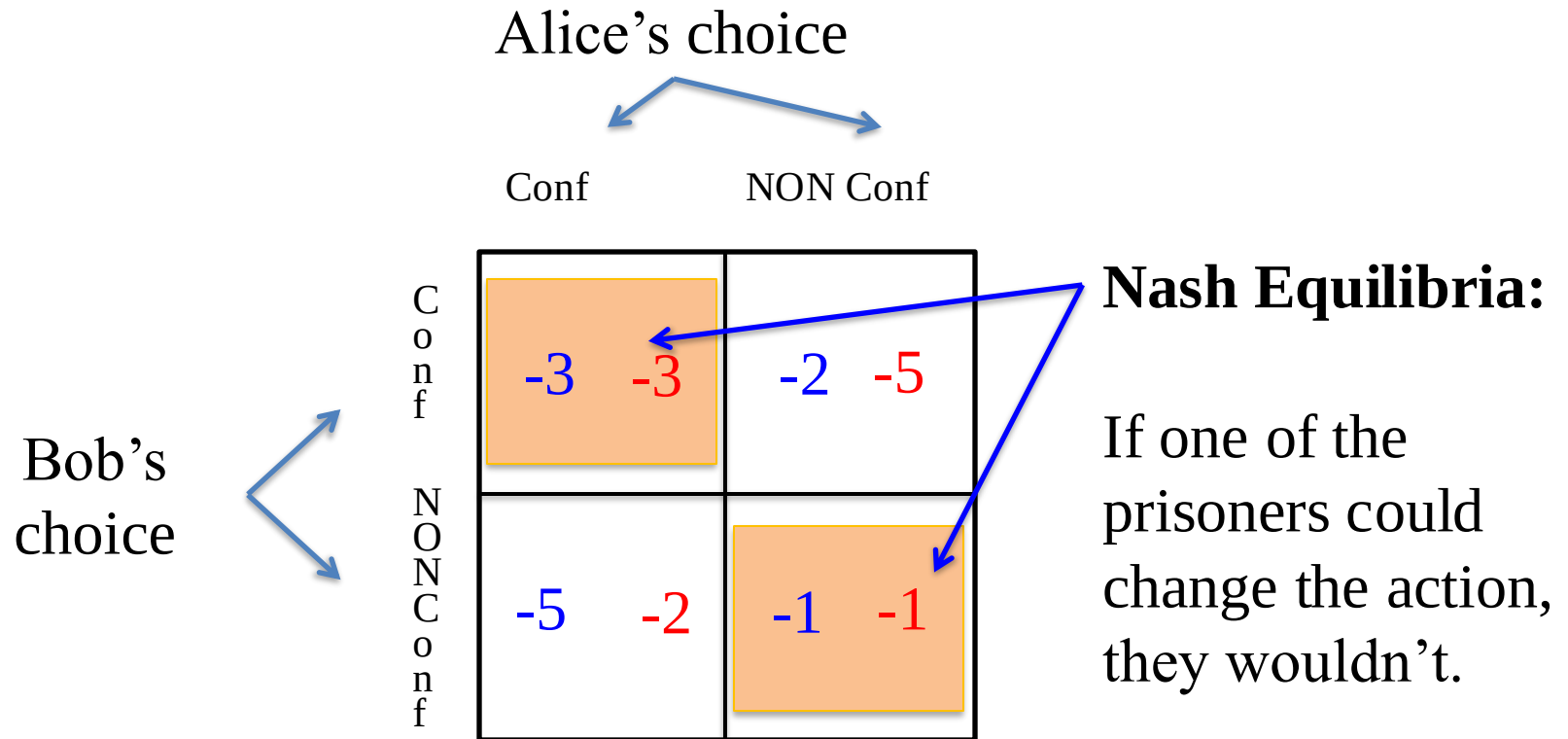
Conf	-3 -3	-2 -5
NON Conf	-5 -2	-1 -1

No more dominant strategy!

Altruistic Prisoner's Dilemma

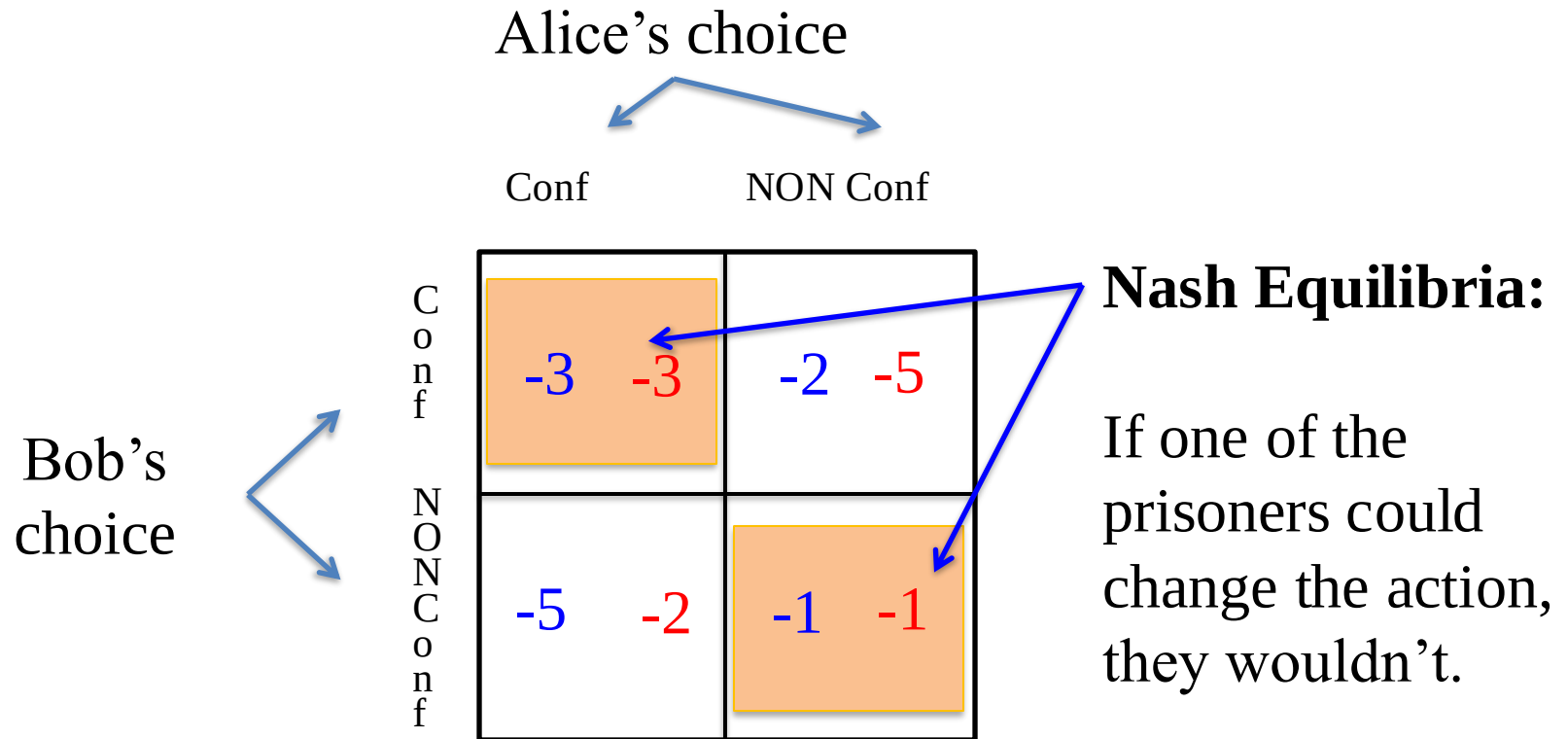


Altruistic Prisoner's Dilemma



However, one of the equilibria is better...

Altruistic Prisoner's Dilemma



However, one of the equilibria is better...

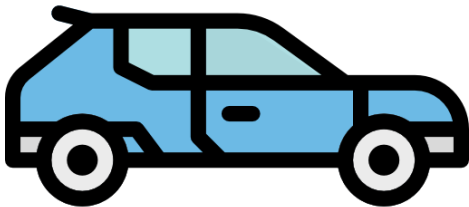
Coordination game

Chicken's game

Two cars drive towards each other.

Every driver can decide whether to swerve or keep going.

- If one swerve while the other doesn't, the first is considered a 'chicken'
- If both don't swerve, they crash
- If both swerve, they'll have a bad reputation among their peers



Chicken's game

- Crash: -10
- Chicken: -1
- Indifference: 0
- Brave: 1

	Swerve	Straight
Swerve	0 0	-1 1
Straight	1 -1	-10 -10

Chicken's game

- Crash: -10
- Chicken: -1
- Indifference: 0
- Brave: 1

	Swerve	Straight
Swerve	0 0	-1 1
Straight	1 -1	-10 -10

Can you compute the equilibria?

Chicken's game

- Crash: -10
- Chicken: -1
- Indifference: 0
- Brave: 1

	Swerve	Straight
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Can you compute the equilibria?

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Anti-Coordination game: each player has a preferred equilibrium

Chicken's game

- Crash: -10
- Chicken: -1
- Indifference: 0
- Brave: 1

	Swerve	Straight
Swerve	0 0	-1 1
Straight	1 -1	-10 -10

Anti-Coordination game: each player has a preferred equilibrium

Best strategy: convince the other you are not going to swerve!

Introducing some math...

Focus of game theory:

Situations (**games**) in which **players** (agents, actors, individuals, decision makers) **interact** with one another, and in which the **outcome** (profit, loss, costs, utility) of each player may depend not just on their own decisions but also on the decisions of the other participants.

How do we make this formal?

Bimatrix games

Prisoner's Dilemma

$$\begin{array}{c}
 P_1 \\
 \begin{array}{c} C \\ NC \end{array}
 \end{array}
 \begin{array}{c}
 P_2 \\
 \begin{array}{c} C \\ NC \end{array}
 \end{array}
 \begin{bmatrix}
 -3, -3 & 0, -5 \\
 -5, 0 & -1, -1
 \end{bmatrix}$$

Bob

		Alice	
		Conf	NON Conf
Bob	Conf	-3 -3	0 -5
	NON Conf	-5 0	-1 -1

Bimatrix games

Prisoner's Dilemma

$$\begin{array}{c} P_1 \\ \begin{array}{c} C \\ NC \end{array} \end{array} \begin{array}{cc} & \begin{array}{c} P_2 \\ \begin{array}{cc} C & NC \end{array} \end{array} \\ \left[\begin{array}{cc} -3, -3 & 0, -5 \\ -5, 0 & -1, -1 \end{array} \right] \end{array}$$

Normal-form description of the game:

{players, their possible actions, outcomes per player for all possible actions}

$$P_1, P_2 \quad \Gamma_1 = \Gamma_2 = \{C, NC\} \quad J_1(C, C) = -3, \dots, J_2(NC, NC) = -1$$

Bimatrix games

Prisoner's Dilemma

		P_2	
		C	NC
P_1	C	$[-3, -3]$	$[0, -5]$
	NC	$[-5, 0]$	$[-1, -1]$

Normal-form description of the game:

{**players**, **their possible actions**, **outcomes per player for all possible actions**}

P_1, P_2 $\Gamma_1 = \Gamma_2 = \{C, NC\}$ $J_1(C, C) = -3, \dots, J_2(NC, NC) = -1$

Strictly dominant strategy: a strategy that is strictly better than others, independently of what the other player does.

When a player has a strictly dominant strategy, we assume that they will play it.

Indeed, C(onfess) is a strictly dominant strategy.

Best response (reply)

Best response (set) R_1 for player P_1 to any action $\gamma_2 \in \Gamma_2$ of player P_2 is defined as:

$$R_1(\gamma_2) = \{\rho \in \Gamma_1 : J_1(\rho, \gamma_2) \geq J_1(\gamma_1, \gamma_2) \forall \gamma_1 \in \Gamma_1\}$$

if each player i maximizes their objective J_i .

If this set a singleton, we Conf about *strict best response*

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If this set a singleton, we Conf about *strict best response*

		P_2	
		L	R
P_1	T	(0,0)	(3,1)
	B	(1,3)	(4,2)

(Profit)

Best response (reply)

Best response (set) R_1 for player P_1 to any action $\gamma_2 \in \Gamma_2$ of player P_2 is defined as:

$$R_1(\gamma_2) = \{\rho \in \Gamma_1 : J_1(\rho, \gamma_2) \geq J_1(\gamma_1, \gamma_2) \forall \gamma_1 \in \Gamma_1\}$$

if each player i maximizes their objective J_i .

If this set is a singleton, we Conf about *strict best response*

		P_2	
		L	R
P_1	T	0,0	3,1
	B	1,3	4,2

(Profit)

Mathematically, denoting the best response of player i by R_i , we can write:

$$R_1(L) = B$$

$$R_1(R) = B$$

$$R_2(T) = R$$

$$R_2(B) = L$$

Best response (reply)

Best response (set) R_1 for player P_1 to any action $\gamma_2 \in \Gamma_2$ of player P_2 is defined as:

$$R_1(\gamma_2) = \{\rho \in \Gamma_1 : J_1(\rho, \gamma_2) \geq J_1(\gamma_1, \gamma_2) \forall \gamma_1 \in \Gamma_1\}$$

if each player i maximizes their objective J_i .

If this set is a singleton, we can talk about *strict best response*

		P_2	
		L	R
P_1	T	0,0	3,1
	B	1,3	4,2
		(Profit)	

Mathematically, denoting the best response of player i by R_i , we can write:

$$R_1(L) = B$$

$$R_1(R) = B$$

$$R_2(T) = R$$

$$R_2(B) = L$$

It appears that the pair of strategies (B, L) is special:
 B is the best response to L and L is the best response to B .

Nash Equilibrium

Even when there are no dominant strategies, we should expect players to use strategies that are best responses to each other.

Suppose that P_1 chooses a strategy γ_1 and P_2 chooses a strategy γ_2 .

This pair of strategies (γ_1, γ_2) is a *Nash equilibrium* if γ_1 is a best response to γ_2 and γ_2 is a best response to γ_1 .

Nash Equilibrium

		P_2	
		L	R
P_1	T	0,0	3,1
	B	1,3	4,2

(Profit)

Note: At Nash equilibrium, there is no unilateral profitable deviation from any of the players involved.

In other words, no player in the game would take a different action as long as every other player remains the same.

Summary so far

- Dominant Strategy
- Best Response
- Nash equilibrium
 - Games with multiple equilibria
 - Choice depends on the strategy and on the ‘temporal dimension’
 - Games without equilibria?

Zero-Sum games

Matching Pennies

- Two Players (A and B)
decide for H or T
- If they get HH or TT,
player A wins
- Otherwise B wins

Zero-Sum games

Matching Pennies

- Two Players (A and B) decide for H or T
- If they get HH or TT, player A wins
- Otherwise B wins

		B		H	T
A	H	1	-1	-1	1
	T	-1	1	1	-1

Zero-Sum games

Matching Pennies

- Two Players (A and B) decide for H or T
- If they get HH or TT, player A wins
- Otherwise B wins

		B	
		H	T
A	H	1 -1	-1 1
	T	-1 1	1 -1

No Nash equilibrium!

Typical in zero-sum games...

Nash's Nobel

How is it that Nash received a Nobel with the following result?

Theorem: *Any game with a finite number of players and a finite number of actions has a mixed-strategy Nash equilibrium.*

Nash's Nobel

How is it that Nash received a Nobel with the following result?

Theorem: *Any game with a finite number of players and a finite number of actions has a mixed-strategy Nash equilibrium.*

- Repeated probabilistic strategies
- In the coin flipping example the equilibrium will be to play H 50% of the time and T 50% of the time

Symmetric games

In a *symmetric game*, every player is identical with respect to the game rules

			
	0,0	-1,1	1,-1
	1,-1	0,0	-1,1
	-1,1	1,-1	0,0

Mixed strategy Nash equilibrium

			
	0	-1	1
	1	0	-1
	-1	1	0

(profit)

Mixed strategy Nash equilibrium

				
	0	-1	1	(profit)
	1	0	-1	
	-1	1	0	

A pair of strategies (now in probabilities) so that each is a best response to the other.

Nash prove that a mixed strategy Nash equilibria exists for any game with a finite number of players and a finite number of strategies.

But how do we find it?

Mixed strategy Nash equilibrium

	p 	q 	$1 - p - q$ 	
	$\begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}$			(profit)
				
				

Let us add probability p to playing rock, probability q to playing paper, and probability $1-p-q$ to play scissors.

Mixed strategy Nash equilibrium

	p 	q 	$1 - p - q$ 	
	$\begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}$			(profit)
				
				

Let us add probability p to playing rock, probability q to playing paper, and probability $1-p-q$ to play scissors.

- Expected payoff from playing rock is
- Expected payoff from playing paper is
- Expected payoff from playing scissors is

Mixed strategy Nash equilibrium

	p 	q 	$1 - p - q$ 	
	$\begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}$	0	-1	1
		1	0	-1
		-1	1	0
				(profit)

Let us add probability p to playing rock, probability q to playing paper, and probability $1-p-q$ to play scissors.

- Expected payoff from playing rock is $0p - 1q + 1(1 - p - q)$
- Expected payoff from playing paper is $1p - 0q - 1(1 - p - q)$
- Expected payoff from playing scissors is $-1p - 1q + 0(1 - p - q)$

If we are at Nash equilibrium, all of these have to be the same.

Mixed strategy Nash equilibrium

	p	q	$1 - p - q$	
				
	$\begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}$			(profit)
				
				

$$\begin{aligned}
 0p - 1q + 1(1 - p - q) &= 1p - 0q - 1(1 - p - q) \\
 1p - 0q - 1(1 - p - q) &= -1p + 1q + 0(1 - p - q)
 \end{aligned}$$

Linear System!
(2 equations in 2 variables)

Mixed strategy Nash equilibrium

	p 	q 	$1 - p - q$ 	
	$\begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}$			(profit)
				
				

$$\begin{aligned}
 0p - 1q + 1(1 - p - q) &= 1p - 0q - 1(1 - p - q) \\
 1p - 0q - 1(1 - p - q) &= -1p + 1q + 0(1 - p - q)
 \end{aligned}$$

Linear System!
(2 equations in 2 variables)

$$p = q = \frac{1}{3}$$

Another bimatrix game

$$\begin{pmatrix} 1,3 & 0,2 \\ 2,-1 & 0,1 \end{pmatrix} \quad (\text{costs})$$

$$A = \begin{pmatrix} 1 & 0 \\ 2 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 3 & 2 \\ -1 & 1 \end{pmatrix}$$

Another bimatrix game

$$A = \begin{pmatrix} 1 & 0 \\ 2 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 3 & 2 \\ -1 & 1 \end{pmatrix}$$

(costs)

- Imagine the row player plays
top row with probability x_1 ,
bottom row with probability $x_2 = 1 - x_1$
- Imagine the column player plays
left column with probability y_1
right column with probability $y_2 = 1 - y_1$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

Another bimatrix game

$$A = \begin{matrix} & \overset{x_1}{\underset{x_2}{}} & \begin{pmatrix} 1 & 0 \\ 2 & 0 \end{pmatrix} \\ & \underset{y_1}{\underset{y_2}} & \end{matrix} \quad \text{(costs)} \quad B = \begin{matrix} & \overset{x_1}{\underset{x_2}{}} & \begin{pmatrix} 3 & 2 \\ -1 & 1 \end{pmatrix} \\ & \underset{y_1}{\underset{y_2}} & \end{matrix}$$

- The expected outcome of Player A is

$$1x_1y_1 + 0x_1y_2 + 2x_2y_1 + 0x_2y_2 = x^T A y$$

- The expected outcome of Player B is

$$3x_1y_1 + 2x_1y_2 + (-1)x_2y_1 + 1x_2y_2 = x^T B y$$

Another bimatrix game

$$A = \begin{matrix} & \begin{matrix} \textcolor{red}{x}_1 & \textcolor{red}{x}_2 \end{matrix} \\ \begin{matrix} \textcolor{green}{y}_1 \\ \textcolor{green}{y}_2 \end{matrix} & \begin{pmatrix} 1 & 0 \\ 2 & 0 \end{pmatrix} \end{matrix} \quad \text{(costs)} \quad B = \begin{matrix} & \begin{matrix} \textcolor{red}{x}_1 & \textcolor{red}{x}_2 \end{matrix} \\ \begin{matrix} \textcolor{green}{y}_1 \\ \textcolor{green}{y}_2 \end{matrix} & \begin{pmatrix} 3 & 2 \\ -1 & 1 \end{pmatrix} \end{matrix}$$

- The expected outcome of Player A is

$$1x_1y_1 + 0x_1y_2 + 2x_2y_1 + 0x_2y_2 = x^T A y$$

- The expected outcome of Player B is

$$3x_1y_1 + 2x_1y_2 + (-1)x_2y_1 + 1x_2y_2 = x^T B y$$

Def. A pair $\textcolor{red}{x}^*, \textcolor{green}{y}^*$ is said to constitute a **(mixed-strategy) Nash equilibrium** solution to a bimatrix game A, B where both players minimize their costs, if

$$\begin{aligned} \textcolor{red}{x}^{*T} A \textcolor{green}{y}^* &\leq \textcolor{red}{x}^T A \textcolor{green}{y}^* \text{ for all possible } \textcolor{red}{x} \\ \textcolor{red}{x}^{*T} B \textcolor{green}{y}^* &\leq \textcolor{red}{x}^{*T} B \textcolor{green}{y} \text{ for all possible } \textcolor{green}{y} \end{aligned}$$

Nash's Nobel

How is it that Nash received a Nobel with the following result?

Theorem: *Any game with a finite number of players and a finite number of actions has a mixed-strategy Nash equilibrium.*

Graphical Method

Bob's loss matrix

Alice's
payoff
matrix

		B plays	
		b_1	b_2
A plays	a_1	1	3
	a_2	4	2

Graphical Method

Alice's payoff matrix

		Bob's loss matrix B plays		
		A plays	b_1	b_2
	a_1	1	3	$1-p$
	a_2	4	2	p

- Alice plays a_2 with prob. p , and a_1 with prob. $1 - p$.

Graphical Method

Alice's payoff matrix

Bob's loss matrix

B plays

A plays

	b_1	b_2	
a_1	1	3	$1-p$
a_2	4	2	p

- Alice plays a_2 with prob. p , and a_1 with prob. $1 - p$.
- What p should Alice pick?

Graphical Method

Alice's payoff matrix

		Bob's loss matrix B plays		
		A plays	b_1	b_2
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- Alice plays a_2 with prob. p , and a_1 with prob. $1 - p$.
- What p should Alice pick?
best guaranteed payoff

Graphical Method

Alice's payoff matrix

		Bob's loss matrix	
		B plays	
A plays	a_1	b_1	b_2
	a_2	b_1	b_2
		1	3
		4	2
		$1-p$	p

- Alice plays a_2 with prob. p , and a_1 with prob. $1 - p$.
- What p should Alice pick?
best guaranteed payoff
 pick p to max the min payoff.

Graphical Method

Alice's payoff matrix

Bob's loss matrix

		B plays		
A plays	a_1	b_1	b_2	
	a_2	1	3	$1-p$
		4	2	p

$1+3p$ $3-p$ summarized payoff

- Alice plays a_2 with prob. p , and a_1 with prob. $1 - p$.
- What p should Alice pick?
best guaranteed payoff
 pick p to max the min payoff.

Graphical Method

Alice's payoff matrix

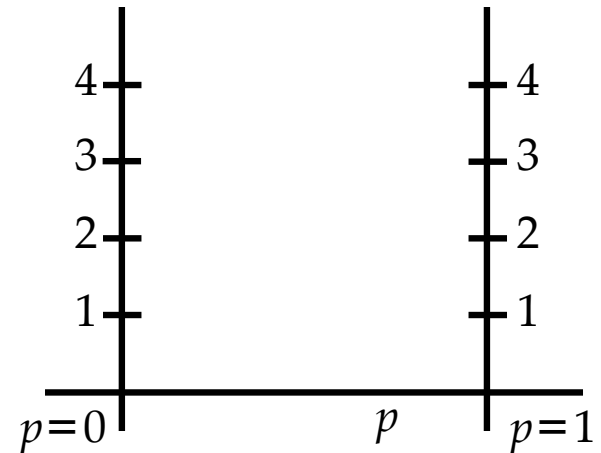
Bob's loss matrix

A plays \ B plays

	b_1	b_2	
a_1	1	3	$1-p$
a_2	4	2	p

$1+3p$ $3-p$ summarized payoff

- Alice plays a_2 with prob. p , and a_1 with prob. $1-p$.
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 pick p to max the min payoff.



Graphical Method

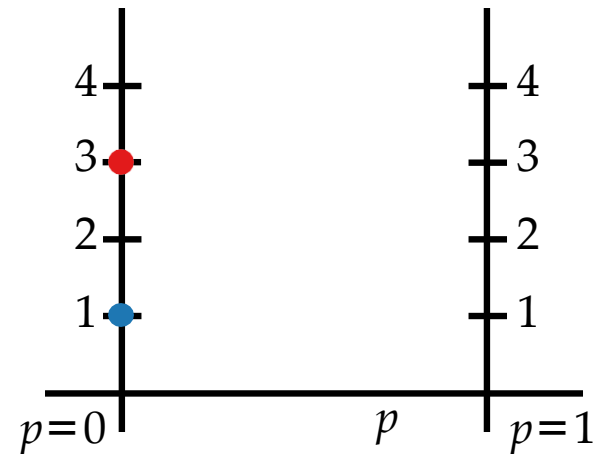
Alice's payoff matrix

Bob's loss matrix

		B plays		
A plays	a_1	b_1	b_2	
	a_2	1	3	$1-p$
		4	2	p

$1+3p$ $3-p$ summarized payoff

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Graphical Method

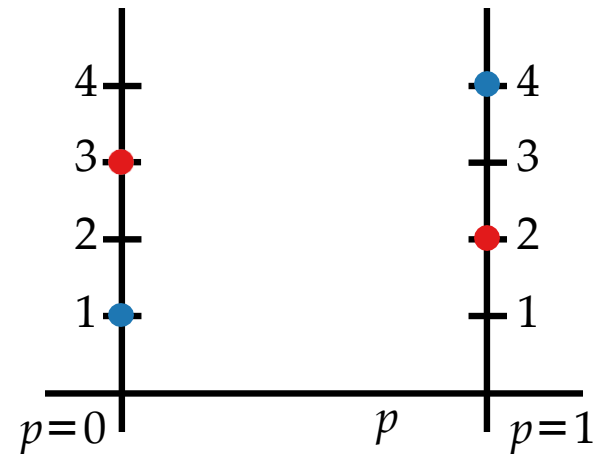
Alice's payoff matrix

Bob's loss matrix

		B plays		
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	a_2	1	3	$1-p$
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$1+3p$ $3-p$ summarized payoff

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- What p should Alice pick?
best guaranteed payoff
 pick p to max the min payoff.



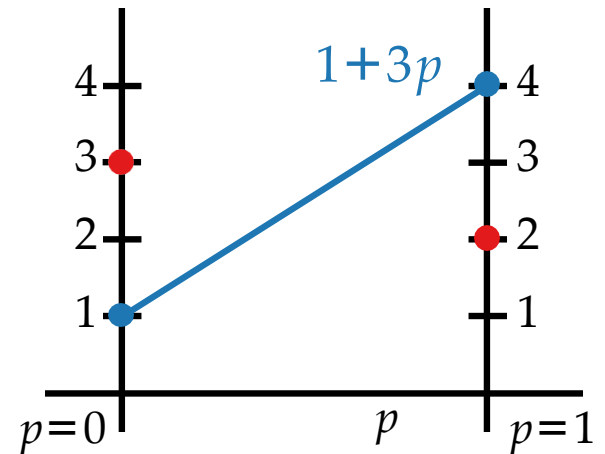
Graphical Method

Alice's payoff matrix

		Bob's loss matrix B plays		
		a_1	a_2	
A plays	b_1	1	4	$1+3p$
	b_2	3	2	$3-p$

summarized payoff

- Alice plays a_2 with prob. p , and a_1 with prob. $1 - p$.
- What p should Alice pick?
best guaranteed payoff
 pick p to max the min payoff.



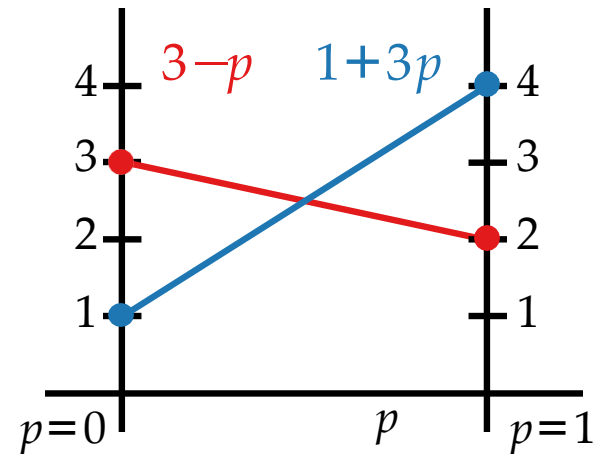
Graphical Method

Alice's payoff matrix

		Bob's loss matrix		
		b_1	b_2	
A plays	a_1	1	3	$1-p$
	a_2	4	2	p

$1+3p$ $3-p$ summarized payoff

- Alice plays a_2 with prob. p , and a_1 with prob. $1-p$.
- What p should Alice pick?
best guaranteed payoff
 pick p to max the min payoff.



Graphical Method

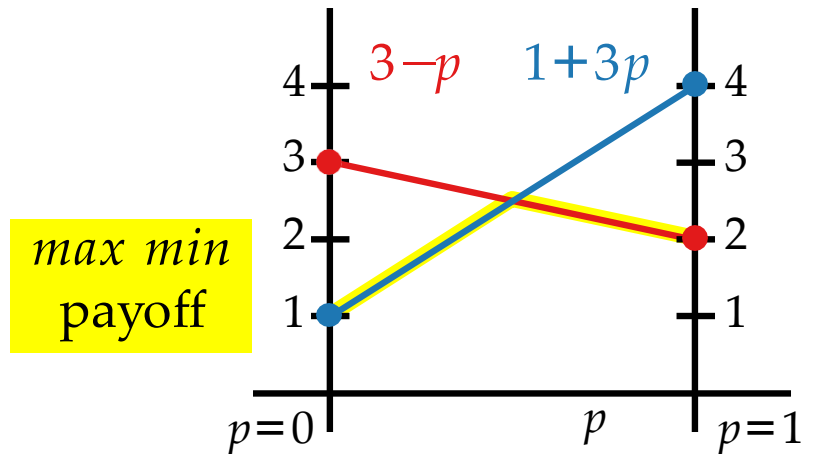
Alice's payoff matrix

Bob's loss matrix

		B plays		
A plays	a_1	b_1	b_2	
	a_2	1	3	$1-p$
		4	2	p

$1+3p$ $3-p$ summarized payoff

- Alice plays a_2 with prob. p , and a_1 with prob. $1-p$.
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best guaranteed payoff
 pick p to max the min payoff.



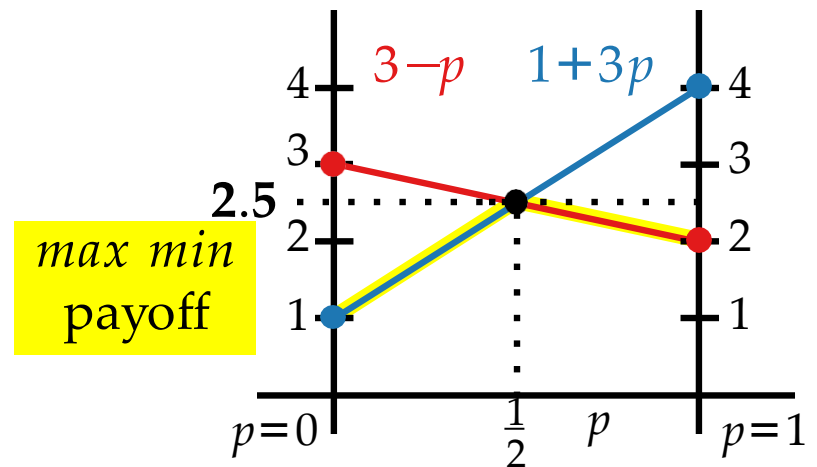
Graphical Method

Alice's payoff matrix

		Bob's loss matrix B plays		
A plays	a_1	b_1	b_2	
		1	3	$1-p$
	a_2	4	2	p

$1+3p$ $3-p$ summarized payoff

- Alice plays a_2 with prob. p , and a_1 with prob. $1-p$.
- What p should Alice pick?
best guaranteed payoff
 pick p to max the min payoff.



max min payoff is 2.5 and
occurs when $p = \frac{1}{2}$

Two-Player, Zero-Sum Games

- What q should Bob pick?

A \ B	b_1	b_2
	$1-q$	q
a_1	1	3
a_2	4	2

Two-Player, Zero-Sum Games

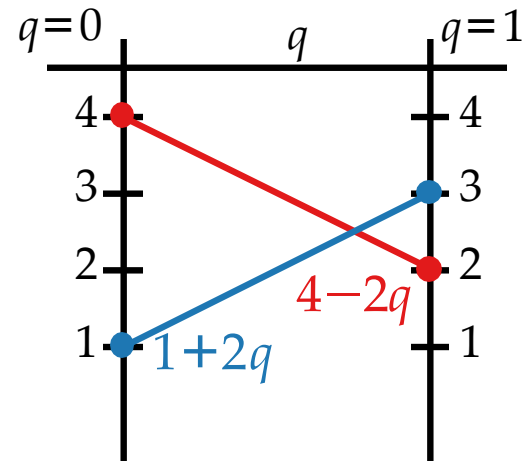
- What q should Bob pick?

		B		
		b_1	b_2	
A	a_1	1	3	$1+2q$
	a_2	4	2	$4-2q$
		$1-q$	q	

Two-Player, Zero-Sum Games

- What q should Bob pick?

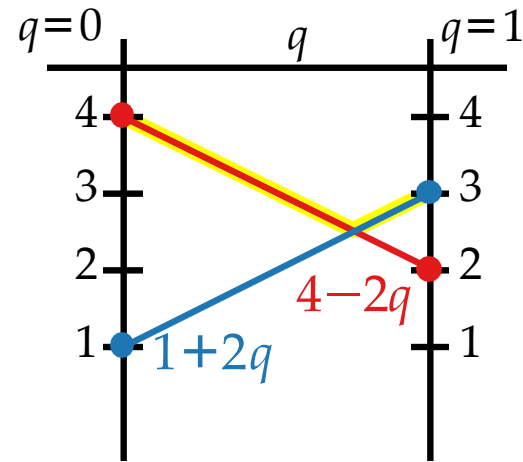
		B		
		b_1	b_2	
A	a_1	1	3	$1+2q$
	a_2	4	2	$4-2q$
		$1-q$	q	



Two-Player, Zero-Sum Games

- What q should Bob pick?

		B		
		b_1	b_2	
A	a_1	1	3	$1+2q$
	a_2	4	2	$4-2q$
		$1-q$	q	

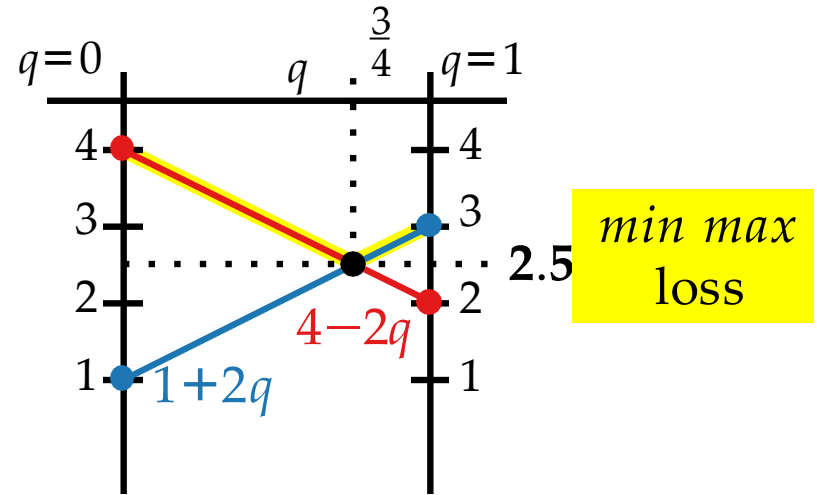


min max
loss

Two-Player, Zero-Sum Games

- What q should Bob pick?

		B		
		b_1	b_2	
A	a_1	1	3	$1+2q$
	a_2	4	2	$4-2q$
		$1-q$	q	



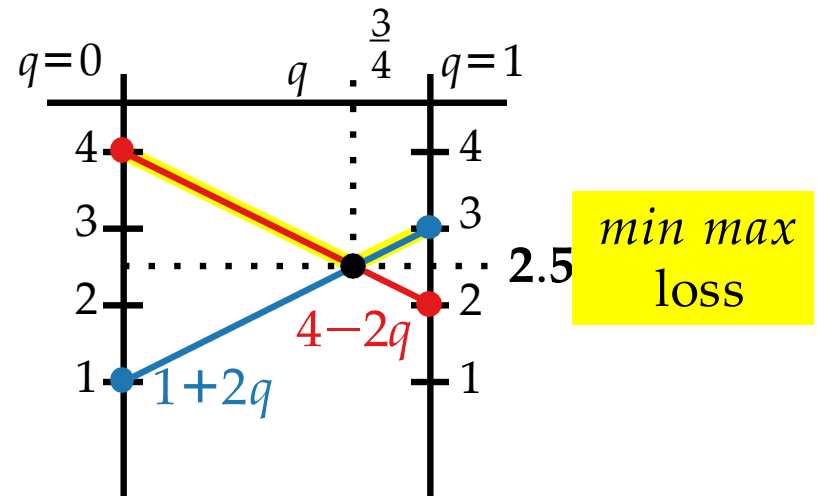
min max loss is 2.5 and
occurs when $q = \frac{3}{4}$

Two-Player, Zero-Sum Games

- What q should Bob pick?

		B		
		b_1	b_2	
A	a_1	1	3	$1+2q$
	a_2	4	2	$4-2q$
		$1-q$	q	

The value is the same!
coincidence?



min max loss is 2.5 and
occurs when $q = \frac{3}{4}$

Two-Player, Zero-Sum Games

Alice's payoff matrix

		Bob's loss matrix	
		B plays	
A plays	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{3}{4}$
	$\frac{1}{2}$	1	3
	$\frac{1}{2}$	4	2

Strategies $(\frac{1}{2}, \frac{1}{2})$ for A, $(\frac{1}{4}, \frac{3}{4})$ for B are

optimal since they both ensure to **maximize the minimum payoff** and (symmetrically) both **minimize the maximum loss**.

The game's **value** is this max min payoff for the first player, here **2.5**.

Two-Player, Zero-Sum Games

Alice's payoff matrix

		Bob's loss matrix	
		B plays	
A plays	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{3}{4}$
	$\frac{1}{2}$	1	3
		4	2

Strategies $(\frac{1}{2}, \frac{1}{2})$ for A, $(\frac{1}{4}, \frac{3}{4})$ for B are

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John von Neumann

[1928] The MiniMax Theorem

Two-Player, Zero-Sum Games

Alice's payoff matrix

		Bob's loss matrix	
		B plays	
A plays	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{3}{4}$
	$\frac{1}{2}$	1	3
		4	2

Strategies $(\frac{1}{2}, \frac{1}{2})$ for A, $(\frac{1}{4}, \frac{3}{4})$ for B are

optimal since they both ensure to **maximize the minimum payoff** and (symmetrically) both **minimize the maximum loss**.

The game's **value** is this max min payoff for the first player, here **2.5**.



John von Neumann

[1928] The MiniMax Theorem

Every $(n \times m)$ matrix S has a **value**, i.e., there is a unique number v with

$$\max_x \min_y xSy^T = v = \min_y \max_x xSy^T$$

row vectors $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^m$

Non-zero Sum Games?

Alice's
payoff
matrix

		B plays	
		b_1	b_2
A plays	a_1	1	3
	a_2	4	2

(max)

Bob's
payoff
matrix

		B plays	
		b_1	b_2
A plays	a_1	6	-4
	a_2	0	5

Non-zero Sum Games?

(A,B) payoff matrix	B plays			
	A plays		b_1	b_2
	a_1		1, 6	3, -4
	a_2		4, 0	2, 5

Non-zero Sum Games?

(A,B) payoff matrix	A plays	B plays	
		b_1	b_2
	a_1	1, 6	3, -4
	a_2	4, 0	2, 5

- Is there a Pure-Strategy Nash equilibrium?

Non-zero Sum Games?

		B plays	
		b_1	b_2
(A,B) payoff matrix	A plays a_1	1, 6	3, -4
	a_2	4, 0	2, 5

- Is there a Pure-Strategy Nash equilibrium?

$$R_A(b_1) = a_2$$

$$R_A(b_2) = a_1$$

$$R_B(a_1) = b_1$$

$$R_B(a_2) = b_2$$

No Pure Equilibrium

Non-zero Sum Games?

		B plays		
		a_1	a_2	
(A,B) payoff matrix	A plays	b_1	b_2	
		$1, 6$	$3, -4$	$1-p$
		$4, 0$	$2, 5$	p
		$1-q$	q	

- Is there a Pure-Strategy Nash equilibrium?
- Is there a Mixed-Strategy Nash equilibrium?

Non-zero Sum Games?

		B plays		
		a_1	a_2	
(A,B) payoff matrix	A plays	b_1	b_2	
		$1-q$	q	
	a_1	1, 6	3, -4	$1-p$
	a_2	4, 0	2, 5	p

- Is there a Pure-Strategy Nash equilibrium?
- Is there a Mixed-Strategy Nash equilibrium?

$$A : \quad 1 - q + 3q = 1 + 2q \quad a_1$$

Non-zero Sum Games?

		B plays		
		a_1	a_2	
(A,B) payoff matrix	A plays	b_1	b_2	
		$1-q$	q	
	a_1	1, 6	3, -4	$1-p$
	a_2	4, 0	2, 5	p

- Is there a Pure-Strategy Nash equilibrium?
- Is there a Mixed-Strategy Nash equilibrium?

$$A : \quad 1 - q + 3q = 1 + 2q \quad a_1$$

$$4(1 - q) + 2q = 4 - 2q \quad a_2$$

$B :$

Non-zero Sum Games?

		B plays		
		a_1	a_2	
(A,B) payoff matrix	A plays	b_1	b_2	
		$1, 6$	$3, -4$	$1-p$
		$4, 0$	$2, 5$	p
		$1-q$	q	

- Is there a Pure-Strategy Nash equilibrium?
- Is there a Mixed-Strategy Nash equilibrium?

$$A : \quad 1 - q + 3q = 1 + 2q \quad a_1$$

$$4(1 - q) + 2q = 4 - 2q \quad a_2$$

$$B : \quad 6(1 - p) \quad b_1$$

$$-4(1 - p) + 5p = -4 + 9p \quad b_2$$

Non-zero Sum Games?

		B plays		
		a_1	a_2	
(A,B) payoff matrix	A plays	b_1	b_2	
		$1-q$	q	
	a_1	1, 6	3, -4	$1-p$
	a_2	4, 0	2, 5	p

- Is there a Pure-Strategy Nash equilibrium?
- Is there a Mixed-Strategy Nash equilibrium?

$$\begin{array}{ll}
 A : & 1 - q + 3q = 1 + 2q \quad a_1 \\
 & 4(1 - q) + 2q = 4 - 2q \quad a_2 \quad 1 + 2q = 4 - 2q \\
 B : & 6(1 - p) \quad b_1 \\
 & -4(1 - p) + 5p = -4 + 9p \quad b_2 \quad 6(1 - p) = -4 + 9p
 \end{array}$$

Non-zero Sum Games?

		B plays		
		a_1	a_2	
(A,B) payoff matrix	A plays	b_1	b_2	
		$1, 6$	$3, -4$	$1-p$
		$4, 0$	$2, 5$	p
		$1-q$	q	

- Is there a Pure-Strategy Nash equilibrium?
- Is there a Mixed-Strategy Nash equilibrium?

$$\begin{array}{ll}
 A : & 1 - q + 3q = 1 + 2q \\
 & 4(1 - q) + 2q = 4 - 2q \\
 B : & 6(1 - p) \\
 & -4(1 - p) + 5p = -4 + 9p
 \end{array}
 \begin{array}{ll}
 a_1 & \\
 a_2 & \\
 b_1 & \\
 b_2 &
 \end{array}
 \begin{array}{l}
 q = \frac{3}{4} \\
 p = \frac{2}{3}
 \end{array}$$

Non-zero Sum Games?

		B plays		
		a_1	a_2	
(A,B) payoff matrix	A plays	b_1	b_2	
		$1-q$	q	
	a_1	1, 6	3, -4	$1-p$
	a_2	4, 0	2, 5	p

- Is there a Pure-Strategy Nash equilibrium?
- Is there a Mixed-Strategy Nash equilibrium?

$$\begin{array}{ll}
 A : & 1 - q + 3q = 1 + 2q \\
 & 4(1 - q) + 2q = 4 - 2q \\
 B : & 6(1 - p) \\
 & -4(1 - p) + 5p = -4 + 9p
 \end{array}
 \begin{array}{ll}
 a_1 & \\
 a_2 & \\
 b_1 & \\
 b_2 &
 \end{array}
 \begin{array}{l}
 q = \frac{3}{4} \\
 p = \frac{2}{3}
 \end{array}
 \begin{array}{l}
 v_A = 2.5 \\
 v_B = 2
 \end{array}$$

Non-zero Sum Games?

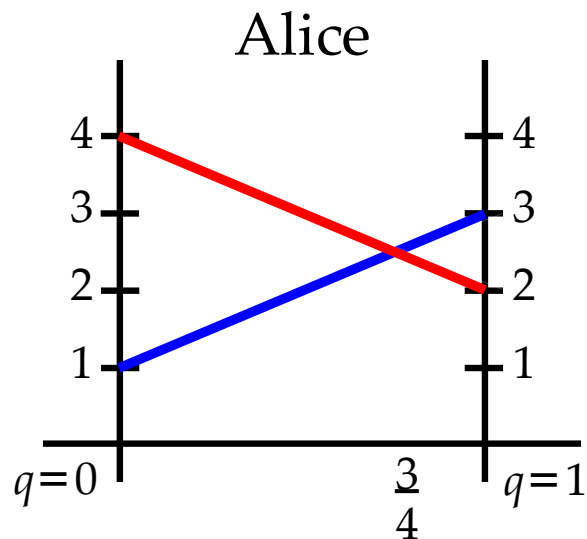
		B plays		
		a_1	a_2	
(A,B) payoff matrix	A plays	b_1	b_2	
		$1, 6$	$3, -4$	$1-p$
		$4, 0$	$2, 5$	p
		$1-q$	q	

- Is there a Pure-Strategy Nash equilibrium?
- Is there a Mixed-Strategy Nash equilibrium?

Non-zero Sum Games?

		B plays		
(A,B) payoff matrix	A plays	b_1	b_2	
	a_1	1, 6	3, -4	$1-p$
	a_2	4, 0	2, 5	p
		$1-q$	q	

- Is there a Pure-Strategy Nash equilibrium?
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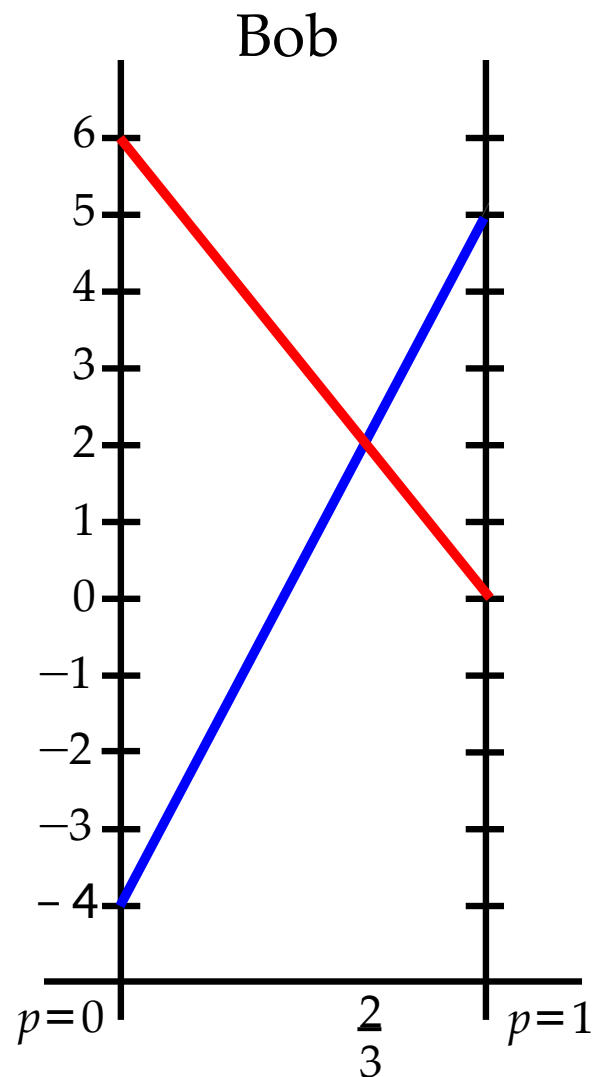
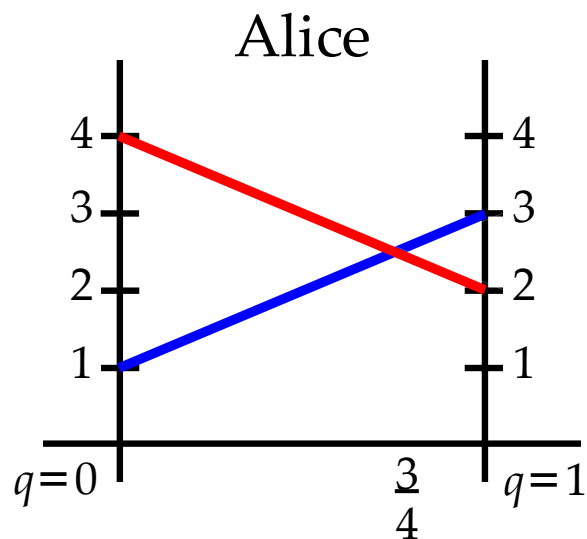


Non-zero Sum Games?

(A,B)
payoff
matrix

		B plays		
		A plays	b_1	b_2
	a_1		1, 6	3, -4
	a_2		4, 0	2, 5
			$1-q$	q
				$1-p$
				p

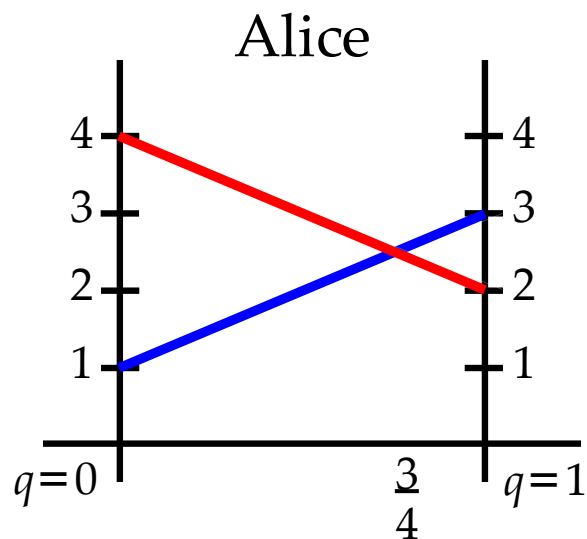
- Is there a Pure-Strategy Nash equilibrium?
- Is there a Mixed-Strategy Nash equilibrium?



Non-zero Sum Games?

(A,B) payoff matrix		B plays		
		A plays		
			b_1 b_2	
	a_1	$\left \begin{array}{cc} 1, 6 & 3, -4 \end{array} \right $	$1-p$	
	a_2	$\left \begin{array}{cc} 4, 0 & 2, 5 \end{array} \right $	p	
		$1-q$ q		

- Is there a Pure-Strategy Nash equilibrium?
- Is there a Mixed-Strategy Nash equilibrium?



$$v_A = 2.5$$

$$v_B = 2$$

